

Product Requirements Document (PRD)

SMK AI-Muhtadin — Official School Website

Version: 1.1
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Status: ☒ Approved — Ready for Development
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1. Executive Summary

Problem Statement

SMK AI-Muhtadin currently lacks a centralized digital presence, making it difficult for prospective students, parents, current students, and the general public to access school information, news, and official announcements in a timely and organized manner.

Proposed Solution

Build a modern, responsive official school website that serves as the single source of truth for school information, featuring a public-facing frontend (React) and a secure content management backend (Laravel + PostgreSQL). The website will be easy to navigate, visually professional, and content-manageable by non-technical staff through an Admin Dashboard (CMS).

Success Criteria (KPIs)

Metric	Target
Page load time (LCP)	≤ 2.5 seconds on a 4G connection
Lighthouse Performance Score	≥ 85
Lighthouse Accessibility Score	100
Mobile responsiveness	100% pages pass on screens ≥ 320px wide
Admin content publish time	≤ 5 minutes from draft to live
MVP launch	Within 1 month of development start

2. User Experience & Functionality

User Personas

Persona	Description	Primary Goal
Prospective Student / Parent	Families considering enrollment at SMK Al-Muhtadin	Find school profile, vision/mission, programs, and contact info
Current Student	Enrolled student looking for news & announcements	Read the latest school news and updates
General Public	Community members, journalists, or institutions	Learn about the school's achievements and activities via news and gallery
School Administrator	Non-technical staff managing content (e.g., teacher or TU staff)	Publish news articles, upload gallery images, update school information

Pages & User Stories

2.1 — Home Page

Story: As a visitor, I want to see a compelling homepage so that I can quickly understand what SMK Al-Muhtadin offers and navigate to what I need.

Acceptance Criteria:

- Displays a hero banner with the school name, tagline, and a clear call-to-action button (e.g., "Lihat Profil Sekolah")
- Shows a "Latest News" section (minimum 3 most recent articles) with thumbnail, title, and date
- Shows a brief "About School" section with a link to the full Profile page
- Includes a footer with school contact info (address, phone, email), social media links, and copyright
- Fully responsive on mobile, tablet, and desktop

2.2 — News Page

Story: As a visitor, I want to read the latest school news and announcements so that I stay informed about events and activities.

Acceptance Criteria:

- Displays a paginated list of news articles (≥ 10 articles per page)
 - Each article card shows: thumbnail image, title, category tag, publication date, and a short excerpt (≤ 150 characters)
 - Clicking a card navigates to a full article detail page
 - Article detail page shows: featured image, title, author, date, body content (rich text / HTML), and related articles
 - News can be filtered/searched by title keyword
 - Supports categories (e.g., "Pengumuman", "Kegiatan", "Prestasi")
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2.3 — Profile Page

Story: As a prospective parent or student, I want to learn about the school's background and leadership so that I can trust in the school's credibility.

Acceptance Criteria:

- Displays school history, founding year, and general description
 - Includes a Principal's message with photo and name
 - Shows a structured list of programs/majors offered (e.g., TKJ, RPL, Akuntansi)
 - Displays school's accreditation status and NPSN (school registration number)
 - Includes a teacher/staff directory section (name, position, photo — optional)
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2.4 — Vision & Mission Page

Story: As a visitor, I want to read the school's vision and mission so that I understand the institution's values and goals.

Acceptance Criteria:

- Clearly separates "Visi" (Vision) and "Misi" (Mission) sections
 - Mission is displayed as a numbered or bulleted list
 - Includes the school's motto if applicable
 - Content is editable via Admin Dashboard
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2.5 — Gallery Page

Story: As a visitor, I want to view photos of school activities so that I can get an impression of campus life and events.

Acceptance Criteria:

- Displays images in a responsive grid layout (3 columns desktop, 2 tablet, 1 mobile)
- Images are grouped by album/category (e.g., "PPDB 2025", "Peringatan HUT RI")
- Clicking an image opens a lightbox with full-size view and navigation (prev/next)
- Albums can be managed (created, renamed, deleted) from the Admin Dashboard
- Images support lazy loading for performance

2.6 — Admin Dashboard (CMS)

Story: As a school administrator, I want to log in to a secure dashboard so that I can manage all website content without needing a developer.

Acceptance Criteria:

- Authentication via email + password with JWT (JSON Web Token)
- Role-based access: `Super Admin` (full access) and `Editor` (can manage news & gallery only)
- **News Management:** Create, Read, Update, Delete (CRUD) articles with a rich-text editor (e.g., TipTap or Quill); upload featured images; set status (Draft / Published)
- **Gallery Management:** Create albums, upload multiple images at once, delete images/albums
- **Profile & Vision/Mission Management:** Edit static page content (school description, vision, mission) via a rich-text editor
- **Dashboard Overview:** Shows counts of total articles, total gallery images, draft articles
- All admin routes protected by authentication middleware
- Audit log: track who published/edited content and when

Non-Goals (Out of Scope for MVP)

The following features are **explicitly excluded** from this version to protect the 1-month timeline:

- ❌ Online Student Enrollment / PPDB Registration Form
- ❌ E-Learning or LMS features (courses, assignments, grades)
- ❌ Student/Teacher Internal Portal (separate login for students)

- ❌ Live chat / chatbot feature
- ❌ Mobile app (iOS/Android)
- ❌ Multi-language support (English/Indonesian toggle)
- ❌ Payment gateway integration

3. Technical Specifications

3.1 Architecture Overview



3.2 Tech Stack

Layer	Technology	Version
Frontend	React (Vite)	React 18+
Frontend UI	Tailwind CSS + shadcn/ui	Latest
Frontend Routing	React Router DOM	v6+
Backend	Laravel	11.x
Authentication	Laravel Sanctum (SPA Token)	Built-in
Database	PostgreSQL	16.x
File Storage	Local Disk (Laravel Storage) → upgrade to S3 in v2	—
Rich Text Editor	TipTap	v2+
Image Optimization	Laravel Intervention/Image	v3
Deployment	VPS (Ubuntu) + Nginx + PHP-FPM	—

3.3 Database Schema (Core Tables)

users

Column	Type	Notes
id	BIGINT (PK)	
name	VARCHAR(255)	
email	VARCHAR(255) UNIQUE	
password	VARCHAR(255)	Bcrypt hashed
role	ENUM('super_admin', 'editor')	
created_at	TIMESTAMP	
updated_at	TIMESTAMP	

news

Column	Type	Notes
id	BIGINT (PK)	
title	VARCHAR(255)	
slug	VARCHAR(255) UNIQUE	URL-friendly
excerpt	TEXT	≤ 150 chars
body	TEXT	HTML/Rich text
featured_image	VARCHAR(500)	File path
status	ENUM('draft', 'published')	
category_id	BIGINT (FK → categories)	
author_id	BIGINT (FK → users)	
published_at	TIMESTAMP NULLABLE	
created_at	TIMESTAMP	
updated_at	TIMESTAMP	

categories

Column	Type	Notes
id	BIGINT (PK)	
name	VARCHAR(100)	e.g., "Pengumuman"
slug	VARCHAR(100) UNIQUE	

albums

Column	Type	Notes
id	BIGINT (PK)	
name	VARCHAR(255)	e.g., "PPDB 2025"
description	TEXT NULLABLE	
created_at	TIMESTAMP	

gallery_images

Column	Type	Notes
id	BIGINT (PK)	
album_id	BIGINT (FK → albums)	
image_path	VARCHAR(500)	
caption	VARCHAR(255) NULLABLE	
order	INT	For sorting
created_at	TIMESTAMP	

pages

Column	Type	Notes
id	BIGINT (PK)	
key	VARCHAR(100) UNIQUE	e.g., "vision_mission", "profile"
title	VARCHAR(255)	
content	TEXT	HTML/Rich text
updated_at	TIMESTAMP	
updated_by	BIGINT (FK → users)	

3.4 API Endpoints (Key)

Public Endpoints (No Auth Required)

Method	Endpoint	Description
GET	/api/news	List published articles (paginated)
GET	/api/news/{slug}	Get single article
GET	/api/categories	List all categories
GET	/api/albums	List all albums with images
GET	/api/pages/{key}	Get static page content (profile, vision)

Admin Endpoints (Auth Required — Bearer Token)

Method	Endpoint	Description
POST	/api/auth/login	Admin login
POST	/api/auth/logout	Admin logout
GET/POST	/api/admin/news	List / Create news
GET/PUT/DELETE	/api/admin/news/{id}	Get / Update / Delete news
POST	/api/admin/news/{id}/publish	Change status to published
GET/POST	/api/admin/albums	List / Create albums
POST	/api/admin/albums/{id}/images	Upload images to album
DELETE	/api/admin/images/{id}	Delete gallery image
PUT	/api/admin/pages/{key}	Update static page content

3.5 Security & Privacy

Concern	Implementation
Authentication	Laravel Sanctum SPA tokens; HTTPS-only cookies
Password Security	Bcrypt hashing (cost factor ≥ 12)
Authorization	Policy-based role checks on every admin endpoint
SQL Injection	Eloquent ORM with parameterized queries
XSS	Input sanitization on rich-text fields (strip scripts); React escapes output by default
CSRF	Laravel CSRF tokens for web forms
File Uploads	Validate MIME types (image only), max 5MB per image, store outside public root
Rate Limiting	Throttle login endpoint: max 5 attempts/minute per IP
Audit Logging	Log all CMS write operations (who, what, when)

4. Risks & Roadmap

4.1 Technical Risks

Risk	Likelihood	Impact	Mitigation
Tight 1-month timeline causes rushed or incomplete features	High	High	Strictly enforce Non-Goals; cut Gallery or Profile to MVP if needed
Non-technical admin struggles with CMS interface	Medium	High	Provide simple onboarding guide/tutorial within the dashboard
Image storage fills up on local VPS	Medium	Medium	Set image size limits (5MB); plan migration to object storage (S3/Wasabi) in v2
No SEO optimization leads to low discoverability	Medium	Medium	Use React Helmet for meta tags; implement sitemap.xml
PostgreSQL unfamiliarity for the team (if used to MySQL)	Low	Medium	Laravel Eloquent abstracts most differences; document any PostgreSQL-specific syntax

4.2 Phased Roadmap

MVP (Month 1) — Core Information Website

- ☐ Project setup: Laravel API + React Vite + PostgreSQL
- ☐ Authentication & Admin Dashboard (login, role management)
- ☐ Home Page (hero, latest news, about snippet, footer)
- ☐ News Module (list, detail, category filter, CRUD admin)
- ☐ Profile Page (static content, editable via CMS)
- ☐ Vision & Mission Page (static content, editable via CMS)
- ☐ Gallery Module (albums, image upload, lightbox, CRUD admin)
- ☐ Deployment to VPS (Nginx + PHP-FPM + Node build)

v1.1 (Month 2–3) — Enrichment

- ☐ Search feature (full-text search on news)
- ☐ Teacher & Staff Directory page
- ☐ Achievements & Awards Showcase section
- ☐ SEO improvements: sitemap.xml, Open Graph meta tags, structured data
- ☐ Image CDN / object storage migration (Wasabi / Cloudflare R2)
- ☐ Analytics integration (Google Analytics 4 or Umami)

v2.0 (Month 4–6) — Advanced Features

- ☐ Online PPDB (Student Enrollment) Registration Form
- ☐ Download Center (forms, documents, syllabus PDF)
- ☐ Academic Calendar with event management
- ☐ Contact Us form with email notification (Laravel Mail + SMTP)
- ☐ Multi-language support (Bahasa Indonesia / English)

5. Resolved Decisions

All open questions have been answered by the stakeholder. No blockers remain for development.

#	Question	Decision
1	Hosting / VPS	✅ School owns an existing VPS — no provisioning needed
2	Domain	✅ Domain is already registered and ready to point
3	Brand Assets	✅ Logo, color palette, and fonts are ready to be handed off
4	Existing Content	✅ Fresh start — no content migration required
5	Email Notifications	❌ Not needed — excluded from all versions
6	Additional Staff Roles	❌ Not needed — only <code>Super Admin</code> and <code>Editor</code> roles required

Infrastructure Summary

Item	Detail
Server	Existing school-owned VPS
OS (recommended)	Ubuntu 22.04 LTS
Web Server	Nginx + PHP 8.3 FPM
Process Manager	Supervisor (Laravel queues)
Database	PostgreSQL 16
Node	Node.js 20 LTS (for React build)
SSL	Let's Encrypt (Certbot)
Domain	Already registered — configure DNS A record to VPS IP

Document finalized and approved by stakeholder on August 9, 2026. Development may commence immediately.